STAT3003: Survey Methods
8. Applied Problems

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*Contains some non-examinable material

Applied Problems

- By now we have learned there are several sources of error for surveys
- Here, we will go through some statistical approaches to dealing with these problems
  - What if the respondent is unlikely to give a truthful answer?
  - What if our sample happened to be not representative of the population?
  - What if non-response is a serious problem?
  - What if the population we study contains elements we are not interested in?

Applied Problems

- Here, we will go through some statistical approaches to dealing with these problems
  - What if the respondent is unlikely to give a truthful answer?
    - Use a random-response model
  - What if our sample happened to be not representative of the population?
    - Use post-stratification
  - What if non-response is a serious problem?
    - Use post-stratification or weighting class adjustments
  - What if the population we study contains elements we are not interested in?
    - Estimate over subpopulations

8.1 Random Response Model

- We have already mentioned how it is natural for respondents to not answer embarrassing/sensitive questions truthfully
- Would you answer these questions honestly in front of a stranger?
  - Have you taken drugs in the past month?
  - Have you ever stolen objects from a shop?
  - Have you copied homework for some CUHK course?

8.1 Random Response Model

- A random-response model uses two questions: one sensitive, one harmless.
- The model prevents the interviewer from knowing which question the respondent has answered
  - This should make the respondent more comfortable and more likely to offer a true answer to the sensitive question
- The random-response model we consider works for yes/no questions
  - Or, more generally, “do you belong to Group A or Group B” type questions

8.1 Random Response Model

- We consider a question that asks whether a person belongs to Group A or Group B
- Group A and Group B partition the population
  - i.e. every person is either in Group A or Group B and not in both
- Let p be the proportion of the population in Group A
- Using a random-response model, we can estimate p without directly asking ‘Are you in Group A?’

8.1 Random Response Model

- The model consists
  - Two questions
    - One is the sensitive question
    - One is a trivial question, for which the probability the respondent answers “Yes” is known
  - A randomization device
    - The respondent uses this to decide which question to answer – the interviewer must not know which
    - We know the probabilities the respondent chooses to answer the sensitive question and the trivial question

8.1 Random Response Model

- Example
  - Sensitive question: have you copied homework for some CUHK course?
  - Trivial question: is the last digit of your phone number in {0,1,2,3,4}?
  - Randomization device: a coin.
    - Heads means the respondent answers the sensitive question
    - Tails means the respondent answers the trivial question
- How can we build an estimator for p from this?

8.1 Random Response Model

- Notation
  - $\phi \coloneqq \Pr(\text{"Yes" answer})$
  - $p \coloneqq \Pr(\text{"Yes" answer}|\text{sensitive question})$
- The law of total probability tells us
  - $\phi = p \times Pr(senstive\ qu.) + \Pr(\text{Yes}|trivial\ qu.) \times \Pr(trivial\ qu.)$
- We know all these probabilities apart from ϕ and p
  - $\Pr(sensitive\ qu.) = \Pr(Heads) = \frac{1}{2} = \Pr(trivial\ qu.)$
  - $\Pr(Yes|trivial\ qu.) = \Pr(phone\ no.ends\ in\ \{0,1,2,3,4,5\}) = \frac{1}{2}$
  - $\Rightarrow \phi = \frac{1}{2}p + \frac{1}{4} \Rightarrow p = 2\phi - \frac{1}{2}$

8.1 Random Response Model

- A sensible choice of estimator would be
  - $\hat{p} = 2\hat{\phi} - \frac{1}{2}$
  - $\widehat{Var}(\hat{p}) = 4\widehat{Var}(\hat{\phi}) = 4\left(1 - \frac{n}{N}\right)\frac{1}{n-1}\,\hat{\phi}(1 - \hat{\phi})$, using the SRS formulae for estimating a population proportion
  - Notice the factor of 4 in the variance
- Activity: use these equations to estimate the number of people in this room who have copied their homework in some CUHK course, and provide a 95% CI

8.1 Random Response Model

- More generally, define
  - $p_S := \Pr(sensitive\ qu.)$
  - $p_{Yes|T} := \Pr(\text{Yes}|\text{trivial qu.})$
- Then we have shown
  - $\phi = pp_S + p_{Yes|T}(1 - p_S)$
  - $p = \frac{\phi - p_{Yes|T}(1-p_S)}{p_S} \Rightarrow \hat{p} := \frac{\hat{\phi} - p_{Yes|T}(1-p_S)}{p_S}$
  - $\widehat{Var}(\hat{p}) = \frac{\widehat{Var}(\hat{\phi})}{p_S^2}$.
  - This shows the penalty for using the random response model: the variance is larger
- We could make p_S larger to reduce this problem, but the respondent may think the interviewer will know which question is being asked

8.2 Post-Stratification

- So far we have assumed that we can place sampling units into strata before we conduct the survey
- But sometimes this is not possible
  - E.g. you want to estimate the mean weight of a population. It makes sense to have two strata: one for men, the other for women
  - You conduct this survey by phone, meaning you not know to which stratum the sampling unit will belong until you speak to them

8.2 Post-Stratification

- Post stratification is stratification after the sample has been selected
- We may post stratify if we feel that our sample does not properly reflect the groupings in the population
- Having conducted the survey, we can post stratify and hopefully have a more accurate estimate

8.2 Post-Stratification

- Consider our weight example
  - We conduct a SRS with $n = 100$
  - We conduct the survey and notice that our sample consisted of 80 women and 20 men
  - Of course, we should expect it to consist of 50 men and 50 women
  - Women are lighter than men, as a results, our sample mean $\bar{y}$ is likely to be too low

8.2 Post-Stratification

- The sample mean is $\bar{y} = 132$. We suspect this is too low. What can we do?
  - We know 50% of the population are men and 50% are women
  - From our data, we see the sample mean for women was $\bar{y}_1 = 120, n_1 = 80$
  - For men, $\bar{y}_2 = 180, n_2 = 20$
  - Why not use $\bar{y}_{pst} \coloneqq 0.5 \times \bar{y}_1 + 0.5 \times \bar{y}_2 = 150$

8.2 Post-Stratification

- More generally, for a population with L strata, with N_i and $\bar{Y}_i$ are the population size and sample mean in strata i respectively, the post-stratified estimator for μ is
  - $\bar{Y}_{pst} = \frac{N_1}{N}\bar{Y}_1 + \frac{N_2}{N}\bar{Y}_2 + \cdots + \frac{N_L}{N}\bar{Y}_L$
  - The formulae for τ and p quickly follow
- Is this exactly the same as the stratified sample mean, $\bar{Y}_{st}$?
  - No.
- Consider the sample sizes for each stratum, $n_1, n_2, \ldots, n_L$. In $\bar{Y}_{st}$ these were decided before we conducted the survey, so they are not random
  - In $\bar{Y}_{pst}$, $n_1, n_2, \ldots, n_L$ were only observed after the survey was conducted – essentially, they are random variables.

8.2 Post-Stratification*

- Let's derive the formula for $Var(\bar{Y}_{pst})$
  - Recall the formula for conditional variance from Chapter 4
  - $Var(\hat{\theta}) = Var(E[\hat{\theta}|s_1]) + E[Var(\hat{\theta}|s_1)]$
- In our case, this becomes
  - $Var(\bar{Y}_{pst}) = Var(E[\bar{Y}_{pst}|n_1, \dots, n_L]) + E[Var(\bar{Y}_{pst}|n_1, \dots, n_L)]$
  - Given $n_1, \dots, n_L$, $\bar{Y}_{pst}$ is unbiased, just like the usual stratified estimator
  - $\Rightarrow E[\bar{Y}_{pst}|n_1, \dots, n_L] = \mu$
  - $Var(\mu) = 0$, hence $Var(\bar{Y}_{pst}) = E[Var(\bar{Y}_{pst}|n_1, \dots, n_L)]$
- We focus on calculating $E[Var(\bar{Y}_{pst}|n_1, \dots, n_L)]$

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8.2 Post-Stratification*

- $Var(\bar{Y}_{pst}|n_1, \dots, n_L)$ is just the usual variance formula for the stratified estimator
  - $Var(\bar{Y}_{pst}|n_1, \dots, n_L) = \sum_{i=1}^{L} \left(\frac{N_i^2}{N^2}\right)\left(\frac{N_i - n_i}{N_i - 1}\right)\left(\frac{\sigma_i^2}{n_i}\right)$
  - $\Rightarrow E\left[Var(\bar{Y}_{pst}|n_1, \dots, n_L)\right] = \sum_{i=1}^{L} \left(\frac{N_i^2}{N^2}\right)\left(\frac{\sigma_i^2}{N_i - 1}\right)\left(N_i E\left[\frac{1}{n_i}\right] - 1\right)$
- Our next task is to work out $E\left[\frac{1}{n_i}\right]$
  - Without loss of generality, let's look at $E\left[\frac{1}{n_1}\right]$

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8.2 Post-Stratification*

- We consider the Taylor series expansion of $f(x) = \frac{1}{x}$ around the point $x = E[n_1]$
  - $f(x) = \frac{1}{E[n_1]} + (x - E[n_1])\left(-\frac{1}{E[n_1]^2}\right) + \frac{1}{2}\,(x - E[n_1])^2\left(\frac{2}{E[n_1]^3}\right) + \cdots$
- Evaluate at $x = n_1$
  - $\frac{1}{n_1} = \frac{1}{E[n_1]} + \frac{(E[n_1] - n_1)}{E[n_1]^2} + \frac{(n_1 - E[n_1])^2}{E[n_1]^3} + \cdots$
- Truncate and take the expectation of both sides
  - $E\left[\frac{1}{n_1}\right] \approx \frac{1}{E[n_1]} + \frac{(E[n_1] - E[n_1])}{E[n_1]^2} + \frac{E[(n_1 - E[n_1])^2}{E[n_1]^3}$
  - $E\left[\frac{1}{n_1}\right] \approx \frac{1}{E[n_1]} + \frac{Var(n_1)}{E[n_1]^3}$

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8.2 Post-Stratification*

- We need to work out $E[n_1]$ and $Var(n_1)$
  - Let p_1 be the proportion of elements in Stratum 1
  - Then $\hat{p}_1 = \frac{n_1}{n} \Rightarrow n_1 = n\hat{p}_1$
  - Hence $E[n_1] = nE[\hat{p}_1] = np_1 = n\frac{N_1}{N}$
  - $Var(n_1) = n^2 Var(\hat{p}_1) = n^2 \frac{N-n}{N-1}\frac{1}{n}\, p_1(1-p_1)$
  - $\Rightarrow Var(n_1) = n\frac{N-n}{N-1}\frac{N_1}{N}\left(1-\frac{N_1}{N}\right)$
- Plug these into our expression for $E\left[\frac{1}{n_1}\right]$

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8.2 Post-Stratification*

- $E\left[\frac{1}{n_1}\right] \approx \frac{1}{E[n_1]} + \frac{Var(n_1)}{E[n_1]^3}$
  - $\Rightarrow E\left[\frac{1}{n_1}\right] \approx \frac{1}{n\frac{N_1}{N}} + \frac{n\frac{N-n}{N-1}\frac{N_1}{N}\left(1-\frac{N_1}{N}\right)}{\left(n\frac{N_1}{N}\right)^3}$
  - $= \frac{N}{nN_1} + \frac{1}{n^2}\frac{N-n}{N-1}\frac{N^2}{N_1^2}\left(1-\frac{N_1}{N}\right)$
- We plug this into $\left(\frac{N_1^2}{N^2}\right)\left(\frac{\sigma_1^2}{N_1-1}\right)\left(N_1 E\left[\frac{1}{n_1}\right] - 1\right)$
  - $= \left(\frac{N_1^2}{N^2}\right)\left(\frac{\sigma_1^2}{N_1-1}\right)\left[\frac{N-n}{n} + \frac{1}{n^2}\frac{N-n}{N-1}\frac{N^2}{N_1}\left(1-\frac{N_1}{N}\right)\right]$
  - $= \frac{N-n}{nN}\frac{N_1}{N}\frac{N_1}{N_1-1}\sigma_1^2 + \frac{1}{n^2}\frac{N-n}{N-1}\left(1-\frac{N_1}{N}\right)\frac{N_1}{N_1-1}\sigma_1^2$

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8.2 Post-Stratification

- Hence $Var(\bar{Y}_{pst}) = E\left[Var(\bar{Y}_{pst}|n_1, \dots, n_L)\right]$
  - $\approx \sum_{i=1}^{L} \frac{N-n}{nN} \frac{N_i}{N} \frac{N_i}{N_i-1} \sigma_i^2 + \sum_{i=1}^{L} \frac{1}{n^2} \frac{N-n}{N-1} \left(1 - \frac{N_i}{N}\right) \frac{N_i}{N_i-1} \sigma_i^2$
  - Recall the unbiased estimator for σ^2 is $\frac{N-1}{N} \hat{\sigma}^2$
- So our choice for estimating the variance is
  - $\Rightarrow \widehat{Var}(\bar{Y}_{pst}) \coloneqq \sum_{i=1}^{L} \frac{N-n}{nN} \frac{N_i}{N} \hat{\sigma}_i^2 + \sum_{i=1}^{L} \frac{1}{n^2} \frac{N-n}{N-1} \left(1 - \frac{N_i}{N}\right) \hat{\sigma}_i^2$
  - The red part is the same as $\widehat{Var}(\bar{Y}_{st})$ if proportional allocation is used
  - The blue part is non-negative and shows the increase in variance that results from the randomness of the $n_1, \dots, n_L$

8.2 Post-Stratification

- So when should we use post stratification, given that it increases the variance of our estimator?
  - The increase in variance is small when n is large (note the $1/n^2$ term in front)
  - We also need the n_i to be relatively large as well, in order for the approximation used to find $\widehat{Var}(\bar{Y}_{pst})$ to work
- Thus we should only use post stratification when n and $n_i, i = 1, \ldots, L$ are large
- As result, we should not post-stratify our sample into too many or too small groups

8.2 Post-Stratification

- Example (Scheaffer et al. (2012))
  - A company knows 40% of its accounts are wholesale and 60% are retail
  - An auditor wishes to sample $n = 100$ accounts to estimate the average amount of money owed
  - An SRS is performed, with these results
    - Wholesale: $n_1 = 70, \bar{y}_1 = 520, s_1 = 210$
    - Retail: $n_2 = 30, \bar{y}_2 = 280, s_2 = 90$
  - Estimate the average amount of money owed to the company, and estimate the variance

8.3 Adjusting for Non-response

- We discussed earlier how non-response is a source of error for your survey
- There are methods to reduce it
  - Callbacks
  - Rewards
  - Incentives
- Here we will consider statistical methods that can reduce the effect of non-response

8.3 Adjusting for Non-response

- Recall post-stratification
  - "We may post stratify if we feel that our sample does not properly reflect the groupings in the population"
- Non-response introduces bias into the survey if
  - Non-respondents differ from the respondents
  - There are enough non-respondents to make these differences significantly affect our estimates
- If our survey has a high non-response rate, our sample may "not properly reflect the groupings in the population"

8.3 Adjusting for Non-response

- We can therefore use post-stratification to adjust our estimates for non-response
- What is the difference between
  - our motivation for post-stratification we discussed earlier and
  - the motivation for post-stratification now?

8.3 Adjusting for Non-response

- Earlier, we were concerned that our simple random sample had chosen an unrepresentative sample by pure chance
  - E.g. 80 women out of a sample of 100 of the general population
- Now, we are concerned that by a higher non-response rate, a section of the population is under-represented
  - Out of a sample of 200 students from a population of 60% female, 40% male, 135 responded. But in the responded sample, 104 were female (77%)

8.3 Adjusting for Non-response

- Example (from Scheaffer et al. (2012))
  – Simple random sample of 200 students from a population of 1500
  – Population is 60% female
  – The purpose of the survey is to estimate the total time (in hours) per week a student devotes to studying
  – 135 responses
  – Of the responses, 104 are female and 31 are male

8.3 Adjusting for Non-response

- We see that, because the male students have a higher non-response rate (lazier?), they are under-represented in our sample
  - Proportion of males in sample = 31/135 = 23%
  - Proportion of males in population = 40%
- Likewise, the female students are over-represented in our sample
  - Proportion of females in sample = 104/135 = 77%
  - Proportion of females in population = 60%

8.3 Adjusting for Non-response

- Thus differing response rates can produce unrepresentative samples
- This is a problem if the responses are likely to vary between the groups
  - Perhaps female students work harder and will therefore respond to the question with higher estimates, and they are more willing to response
  - Perhaps male students work less and will therefore respond to the question with lower estimates, and they are less willing to response

8.3 Adjusting for Non-response

- If the responses do not vary between the groups, there isn't a problem
- Example
  - Simple random sample of 200 students from a population of 1500
  - Population is 80% right-handed
  - Question 'Estimate the total time (in hours) per week a student devotes to studying'
  - 135 responses
  - Of the responses, 80 are right-handed and 55 are left-handed ⇒ at 59% of the sample, right-handers are under-represented. Is this a problem?
- No, because total time per week is unlikely to vary between the Left/right-handed groups

8.3 Adjusting for Non-response

- Back to the female/male grouping:
- Say we naively estimate the mean number of hours a week spent on studying, using the SRS estimates. We find the following
  - $\bar{y} = 11.378$
  - $\widehat{Var}(\bar{Y}) = 0.301 = 0.548^2$
  - Then we decide to post-stratify, using our formulas from before

8.3 Adjusting for Non-response

post-stratification adjustment

- We work out that the sample mean and s.d. for the women in the survey is $\bar{y}_f = 12.375$, $s_f = 6.676$
- And for the men in the sample, it is $\bar{y}_m = 8.032, s_m = 5.596$
- We estimate the mean number of hours per week using
  - $\bar{Y}_{pst} = \frac{N_1}{N}\bar{Y}_1 + \frac{N_2}{N}\bar{Y}_2 + \cdots + \frac{N_L}{N}\bar{Y}_L$
  - $\bar{y}_{pst} = \frac{900}{1500} \times 12.375 + \frac{600}{1500} \times 8.032 = 10.638$
- And the variance/s.d. of our estimate is
  - $\widehat{Var}(\bar{Y}_{pst}) = \frac{1}{n}\left(1 - \frac{n}{N}\right)\sum_{i=1}^{L}\frac{N_i}{N}\hat{\sigma}_i^2 + \frac{1}{n^2}\frac{N-n}{N-1}\sum_{i=1}^{L}\left(1 - \frac{N_i}{N}\right)\hat{\sigma}_i^2$
  - $\widehat{Var}(\bar{Y}_{pst}) = 0.2647 + 0.0018 = 0.267, \sqrt{\hat{V}(\bar{y}_{pst})} = 0.516$

8.3 Adjusting for Non-response

post-stratification adjustment

- Notice that our post-stratified estimate is slightly lower, as it should be to adjust the low number of males in the survey
- And the standard deviation has reduced also, because we have taken advantage of the small variance within groups
- This kind of adjustment for the non-response is called a post-stratification adjustment

8.3 Adjusting for Non-response

weight-class adjustment

- Notice that in this case, we knew $\frac{N_f}{N}$ and $\frac{N_m}{N}$
- Consequently we could use the standard post-stratification method to adjust (hence that method's name)
- What if we don't know $\frac{N_i}{N}$ a priori (before)?
- We estimate them from the whole sample, i.e. including the non-respondents
  - This method is called a weight-class adjustment

8.3 Adjusting for Non-response

weight-class adjustment

- This requires us having some knowledge about the non-respondents
  - E.g. whether they are male or female
- Out of the original sample of 200, say 130 were female students and 70 were male
  - We can use this information to estimate $\frac{N_f}{N}$ and $\frac{N_m}{N}$
- Sensible estimates for N_f and N_m are
  - $\widehat{N}_f = N \times \frac{130}{200} = 975$ and $\widehat{N}_m = N \times \frac{70}{200} = 525$

8.3 Adjusting for Non-response

weight-class adjustment

- We use these $\hat{N}_f$ and $\hat{N}_m$ in the post-stratification formulae i.e.
  - $\bar{y}_{wc} = \frac{975}{1500} \times 12.375 + \frac{525}{1500} \times 8.0323 = 10.855$
- And the variance/standard deviation of the estimate (again, using the post-stratification formulas, but only an approximation due to the further randomness) is
  - 0.271 and 0.521 respectively
- It looks like this method, called weight-class adjustment, has even lower variance
- Be careful, this estimate is biased because of the estimation of the stratum sizes.

8.4 Subpopulations

- We have talked about the problem of coverage
  - When our frame does not completely cover the population of interest
- What if we have the opposite problem?
  - Our frame lists more than the population we are interested in
- Perhaps we are interested in a subpopulation
  - Subpopulations are sometimes called “domains” or “subdomains”

8.4 Subpopulations

- Examples
  - Frame: a list of households in a village, but our target population is all households in the village with children
  - Frame: all people in a district, but our target population is all the registered voters in the district
  - Frame: a list of all CUHK students, but our target population is all the students who live on-campus

8.4 Subpopulations

- In cases like these, we actually want to estimate the parameters of a subpopulation, not the population
- But due to the sampling method, we do not know whether a respondent/observation is a member of the subpopulation until after we have sampled it
- That is, we cannot tell *a priori* whether an element is in the subpopulation or not

8.4 Subpopulations

- How can we estimate parameters over a subpopulation?
- Notation
  - N is the population size
  - N_1 is the subpopulation size
  - μ_1 is the subpopulation mean; $\mu_1 = \frac{1}{N_1}\sum_{j=1}^{N_1} Y_{1j}$
  - n is the sample size of the SRS drawn from the whole population
  - n_1 is the number of sampled elements in the subpopulation
  - Y_{1i} is the measurement of the ith element in the sample from the subpopulation.

8.4 Subpopulations

- Thus a sensible estimator for μ_1 is
  - $\hat{\mu}_1 := \bar{Y}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} Y_{1i}$
- This looks like a standard, unbiased sample mean, except
  - n_1 is random
  - $\bar{Y}_1$ is actually a ratio estimator
- For each element i in the sample, we measure two things
  - Y_i, the response of interest
  - $X_i = 1$ if element i is in the subpopulation, $X_i = 0$ otherwise
  - defining $U_i = Y_i X_i$, then $\bar{Y}_1 = \frac{\sum_{i=1}^{n} Y_i X_i}{\sum_{i=1}^{n} X_i} = \frac{\sum_{i=1}^{n} U_i}{\sum_{i=1}^{n} X_i} = \hat{R}$, a standard estimator of the ratio
- Lecture 6 provides formulae for estimating variance
  - $\widehat{Var}(\bar{Y}_1) \approx \frac{1}{\mu_X^2}\left(1 - \frac{n}{N}\right)\frac{1}{n}\frac{1}{n-1}\sum_{i=1}^{n}\left(U_i - \hat{R}X_i\right)^2$

8.4 Subpopulations

- This can be simplified. Note
  - $\left(U_i - \hat{R}X_i\right)^2 = X_i^2\left(Y_i - \hat{R}\right)^2 = X_i^2(Y_i - \bar{Y}_1)^2$
  - $\Rightarrow \sum_{i=1}^{n}\left(U_i - \hat{R}X_i\right)^2 = \sum_{j=1}^{n_1}\left(Y_{1j} - \bar{Y}_1\right)^2 = (n_1 - 1)\hat{\sigma}_1^2$, where $\hat{\sigma}_1^2$ is the sample variance of observations from the subpopulation
  - $\Rightarrow \widehat{Var}(\bar{Y}_1) = \frac{1}{\mu_X^2}\left(1 - \frac{n}{N}\right)\frac{1}{n}\frac{n_1-1}{n-1}\hat{\sigma}_1^2$
- Note $\mu_X = \frac{\tau_X}{N} = \frac{N_1}{N}$
  - $\Rightarrow \widehat{Var}(\bar{Y}_1) = \frac{N^2}{N_1^2}\left(1 - \frac{n}{N}\right)\frac{1}{n}\frac{n_1-1}{n-1}\hat{\sigma}_1^2$

8.4 Subpopulations

- The point estimate $\bar{Y}_1$ doesn't require us to know N_1, but $\widehat{Var}(\bar{Y}_1)$ does
  - If $\frac{N_1}{N}$ is unknown, we plug in $\frac{n_1}{n}$ in its place
  - $\Rightarrow \widehat{Var}(\bar{Y}_1) \approx \left(1 - \frac{n}{N}\right)\left(\frac{n}{n-1}\right)\frac{n_1 - 1}{n_1}\frac{1}{n_1}\hat{\sigma}_1^2$
  - And if n and n_1 are large, $\widehat{Var}(\bar{Y}_1) \approx \left(1 - \frac{n}{N}\right)\frac{\hat{\sigma}_1^2}{n_1}$
  - Which looks a little bit like the usual SRS formula

8.4 Subpopulations

- Estimating the subpopulation total is similar to before
  - Let τ_1 be the subpopulation total
  - Then we may estimate τ_1 by $N_1\bar{Y}_1$
  - $\widehat{Var}(N_1\bar{Y}_1) = N_1^2\widehat{Var}(\bar{Y}_1)$
- Again, this estimate requires us knowing N_1
  - What if do not know N_1 and we want to estimate τ_1?
  - Can we use an alternate estimator?

8.4 Subpopulations

- Recall variable $U_i = Y_i X_i$
  - So $U_i = Y_i$ if element i is in the subpopulation, or $U_i = 0$ otherwise
  - Then $\tau_1 = \sum_{j=1}^{N} u_j$, where $u_j = 0$ if element j is not in the subpopulation, and equals the response of element j if it is in the subpopulation
- This suggests we use an ordinary SRS estimator for τ_1
  - $\hat{\tau}_1 = \frac{N}{n} \sum_{i=1}^{n} U_i$, $\widehat{Var}(\hat{\tau}_1) = N^2 \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}_u^2}{n}$
  - Where $\hat{\sigma}_u^2$ is the sample variance of the U_is
  - Note we don't need to know (or estimate) N_1 at all

8.4 Subpopulations

- Example
  - We want to estimate the total weekly amount spent on food by families with children in a village
  - We have a list of all the families in the village and this is our frame
  - But some families do not have children
  - $N = 250, n = 50, N_1$ is unknown
  - $n_1 = 42, \sum_{j=1}^{42} y_{1j} = 1720, s_1^2 = 42.975$

8.4 Subpopulations

- Thus our estimate for the total amount spend on food per week by families in the village with children is

  $-\frac{N}{n}\sum_{j=1}^{n_1} y_{1j} = \frac{250}{50} \times 1720 = 8600$

- How can we calculate the $\hat{\sigma}_u^2$ required to estimate the variance?

8.4 Subpopulations

- Remember that
  - $\hat{\sigma}_u^2 = \frac{1}{n-1}\sum_{i=1}^{n}(U_i - \overline{U})^2$
  - $= \frac{1}{n-1}\left(\sum_{i=1}^{n} U_i^2 - n(\overline{U})^2\right)$
  - We need to calculate $\overline{U}$ and $\sum_{i=1}^{n} U_i^2$
- Calculating $\overline{U}$ is simple enough
  - $\sum_{i=1}^{n} U_i = \sum_{j=1}^{42} y_{1j} + 8 \times 0 = 1720$
  - The sample size is 50 $\Rightarrow \bar{u}$ in this case is $\frac{1720}{50} = 34.4$

8.4 Subpopulations

- Now to find $\sum_{i=1}^{n} U_i^2$
  - Remember we know s_1^2
  - $s_1^2 = \frac{1}{n_1-1}\left(\sum_{j=1}^{n_1} y_{j1}^2 - n_1(\bar{y}_1)^2\right) = 42.975$
  - We know $\bar{y}_1 = \frac{1720}{42} = 40.95$
  - $\Rightarrow 42.975 = \frac{1}{42-1}\left(\sum_{j=1}^{n_1} y_{j1}^2 - 42 \times 40.95^2\right)$
  - $\sum_{j=1}^{n_1} y_{j1}^2 = 72191.88$
  - $\sum_{i=1}^{n} U_i^2 = \sum_{j=1}^{42} y_{j1}^2 + 8 \times 0^2 = 72191.88$

8.4 Subpopulations

- Hence $s_u^2 = \frac{1}{50-1} \times (72191.88 - 50 \times 34.4^2) = 265.79$
- Hence
  - $\widehat{Var}(\hat{\tau}_1) = 250^2 \left(1 - \frac{50}{250}\right)\left(\frac{265.79}{50}\right) = 265{,}790$
- We now have two estimators for τ_1
  - $N_1 \bar{Y}_1$, a ratio estimator
  - $N\bar{U}$, an unbiased estimator
- In general, which has the greater variance?

8.4 Subpopulations

- We revisit the survey to estimate the total amount of money families with children in a village spend per week on food
  - $N = 250, n = 50, n_1 = 42, \sum_{j=1}^{42} y_{1j} = 1720, s_1^2 = 42.975$
  - Now we also know $N_1 = 210$
  - We have already calculated $\widehat{Var}(N\bar{U}) = 515.55^2$
  - $N_1\bar{Y}_1$ has estimated variance
    - $N_1^2\widehat{Var}(\bar{Y}_1) = N_1^2 \times \frac{N^2}{N_1^2}\left(1 - \frac{n}{N}\right)\frac{1}{n}\frac{n_1-1}{n-1}\hat{\sigma}_1^2 = 189.63^2$
    - This is much lower. Why?

8.4 Subpopulations

- Let's compare the variances
  - $\frac{\widehat{Var}(N\bar{U})}{N_1^2\widehat{Var}(\bar{Y}_1)} = \frac{N^2\left(1-\frac{n}{N}\right)\frac{\hat{\sigma}_u^2}{n}}{N^2\left(1-\frac{n}{N}\right)\frac{1}{n}\frac{n_1-1}{n-1}\hat{\sigma}_1^2} = \frac{n-1}{n_1-1}\frac{\hat{\sigma}_u^2}{\hat{\sigma}_1^2}$
  - Note $n > n_1$
  - $\hat{\sigma}_1^2$ is the sample variance of a sample like $\{y_{11}, y_{12}, \dots, y_{1n_1}\}$
  - $\hat{\sigma}_u^2$ is the sample variance of $\{y_{11}, y_{12}, \dots, y_{1n_1}, 0,0, \dots, 0\}$, with $n - n_1$ zeroes
  - Usually we expect $\hat{\sigma}_u^2 > \hat{\sigma}_1^2$